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NO.	NAME	STUDENT ID NO.	SIGNATURE	DATE
1.	Maryam Laisha Mumthaz	25132291		12/06/26

2.	Kathryn Ann Villacrusis Concepcion	17093683		12/06/26
3.	Angeline Kojansow	25141730		12/06/26
4.	Thighan Dinesh Devagar	18042689		12/06/26

-

E-mail Address / Addresses (according to the order of names above):

1. 25132291@imail.sunway.edu.my	4. 18042689@imail.sunway.edu.my
2. 17093683@imail.sunway.edu.my	5.
3. 25141730@imail.sunway.edu.my	6.

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[] I / We used the following A.I. tools to produce content in this submission:

Tool	Purpose	Prompts	Sections where AI output was used / Outcome(s) in the submission
<i>Gemini</i>	<i>Used to assist with Python programming in Jupyter Notebook, such as troubleshooting code errors.</i>	<i>"Why am I getting this Python error?" "How to solve the error?"</i>	<i>Python Program Development</i>
<i>Claude</i>	<i>Used to provide guidance and tutorials on Power BI development and visualization best practices.</i>	<i>"How can I build a gauge chart for customer churn?" "Explain how to create a calculated measure using DAX."</i>	<i>Dashboard Development on PowerBI</i>

ChatGPT	Used to assist with report organization, section structure, content refinement, and academic writing improvements. Suggestions were adapted and reviewed by the team.	"How should I structure a Business Intelligence report?" "How can I improve the flow between these sections?" "Reduce repetition in this paragraph while maintaining academic tone."	Report Structure Language Editing and Proofreading
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1. Abstract & Introduction

1.1 Abstract

In this study, we have created a Business Intelligence (BI) solution utilizing Power BI for the Brazilian e-commerce platform Olist, in order to tackle several issues such as fragmented data, poor visibility into operational performance, along with a significant need for a centralized environment in order to enable data driven decision making. As such, two different datasets from within Olist's public repository have been combined with an AI generated customer behavior dataset, by utilizing a hybrid ETL / ELT procedure. In addition to this, Python was utilized for cleaning the data, preprocessing as well as feature engineering, while Power BI was mainly utilized for the data modelling aspects along with the dashboard construction. In order to enable this cross functional analysis, a Galaxy Schema was utilized to organize the combined data.

The BI solution was mainly built around five distinct objectives which are as follows:

- Sales and Revenue Analysis
- Delivery and Customer Satisfaction Monitoring
- Seller Performance Evaluation
- Marketing Funnel Effectiveness Assessment
- Customer Segmentation - CLV & Churn Risk Analysis.

In order to meet these objectives and to aid in the interactive reporting and business analysis, a set of KPIs and DAX measurements were developed. Several significant business insights were brought to light with the help of this analysis. First and foremost, there was a relatively high increase in revenue, although the payment activity was mainly centered around credit card payments. Second, even though the delivery performance was mostly positive, it was discovered that there are several regional differences in logistical efficiency. Moreover, the seller performance analysis revealed that a small number of top tier sellers accounted for the majority of the total income, while the marketing analysis revealed that there are a considerable amount of disparities when it comes to conversion effectiveness between acquisition channels. Furthermore, the customer analysis highlighted that there were significant differences in CLV, churn risk, and purchasing behavior among customer segments.

Lastly, four different interactive dashboards have been created in order to assist with Olist's strategic and operational decision making process: an Executive Business Performance Dashboard, an Operational Performance & Lead Conversion Dashboard, a Customer Intelligence

& Predictive Analytics Dashboard, and finally a Logistics & Delivery Performance Dashboard. The findings illustrate how these integrated analytics can translate operational data into meaningful business insights, which ultimately results in better logistical performance, seller management, marketing effectiveness as well as customer retention

1.2 Business Domain Overview

In recent years, the e-commerce industry has expanded a considerable amount, most likely due to changing customer shopping behaviors, internet becoming more and more accessible and mobile commerce acceptance. Global e-commerce sales surpassed USD 5.8 trillion in 2023 and are predicted to reach USD 8 trillion by 2027, according to (Statista, 2024). Due to their quick growth, digital markets are now data intensive business settings that produce massive amounts of operational, transactional, and customer data.

With online retail sales expected to reach over USD 34 billion in 2023, Brazil is one of the biggest e-commerce markets in Latin America (eMarketer, 2024). Due to the industry's fierce rivalry, wide range of consumer tastes, and globally dispersed logistics networks, businesses must constantly assess their operational effectiveness, customer behavior, and sales performance.

Through a centralized platform, Olist, links big online marketplaces with small and medium sized businesses, giving merchants access to a larger client base, Olist enables end to end e-commerce operations across various corporate areas by connecting sellers, consumers, products, orders, payments, and logistical services. Large datasets about sales transactions, product details, customer feedback, vendor activity, payment methods, and delivery performance are produced by the platform (Braga et al., 2019).

Business decisions rely on efficient collection, integration and analysis of data due to the very data rich environment created by marketplace operations. According to research, businesses that operate in these kinds of settings can benefit greatly from analytics and business intelligence skills that turn unprocessed data into useful insights for strategic and operational decision making (Wamba et al., 2015). As a result, the e-commerce industry offers a perfect setting for implementing business intelligence solutions to assist customer analytics, sales optimization, logistics management, performance monitoring and business forecasting.

Organizations are depending more and more on data driven strategies to boost customer satisfaction, boost competitiveness, and foster long term company success as the amount and complexity of marketplace data continue to increase. Businesses may combine data from many

sources, depict important performance indicators, and support well informed decision making at various management levels with the help of business intelligence tools like Power BI.

The e-commerce domain was selected for this project because it generates large volumes of interconnected transactional, customer, logistics, seller, and marketing data that are highly suitable for Business Intelligence analysis. The Olist marketplace dataset provides a realistic business environment with multiple related datasets that can be integrated through common identifiers, making it appropriate for demonstrating data modelling, ETL/ELT processes, KPI development, predictive analytics, and dashboard-based decision support. Furthermore, the availability of sales, delivery, customer satisfaction, seller performance, and marketing information enables the development of a comprehensive BI solution that addresses both operational and strategic business challenges within a single analytical platform.

1.3 Problem Statement

Businesses that operate in dynamic e-commerce settings produce enormous amounts of logistical, behavioral and transactional data (Sharda et al., 2021). Nevertheless, until data is successfully processed, integrated and visualized to provide support for managerial choices, its simple availability does not fully translate into value for business (Sharda et al., 2021). There are several significant business information gaps that have been found in the Olist case:

Firstly, the datasets are disconnected. Without data being manually aggregated, transactional, seller, product and customer data are kept in distinct databases without a single reporting view, leading to comprehensive performance analysis being impossible. Secondly, there are several blind spots in the delivery performance. Olist's management faces several issues in finding inefficient carriers and poor delivery routes due to the fact that they are heavily reliant on third party logistics companies, meaning that they have a significant lack of integrated delivery tracking KPIs (Christopher, 2016).

Additionally, they also don't have proper seller performance benchmarks. Olist consists of around 90,000 registered sellers, however, they don't have a common mechanism for measuring their income contribution, their product quality ratings, or their order fulfillment performance. Moreover, they also face the issue of insufficient use of CLV analysis. Olist's strategies for customer retention and loyalty are considerably limited as they don't utilize data driven segmentation which is based on customer value and purchase behavior (Kumar & Reinarts, 2018). Similarly, their predictive skills are also quite restricted, considering their current reporting is

mostly descriptive and mainly focuses on historical performance rather than revenue trends, seasonal demand patterns, or product category performance.

These difficulties are likely to lead to operational inefficiencies and missed strategic possibilities. A BI solution developed in Power BI can address these gaps by integrating data from multiple sources, establishing analytical KPIs, and providing interactive dashboards that support evidence-based decision-making across different organizational levels (Sharda et al., 2021).

1.4 Business Objective

The main objective of the project is to use Power BI to create an integrated business intelligence system that converts unprocessed marketing as well as e-commerce data into useful insights for predictive, strategic and operational decision making. The solution is intended to give relevant stakeholders a thorough understanding of business performance in terms of consumer behavior, logistical operations, sales, seller performance and marketing efficacy. Five major business objectives were set in order to accomplish this.

BO1: Sales & Revenue

The analysis will employ important data like total orders, items sold, Gross Merchandise Value (GMV), year-to-date revenue, and year-over-year increase in order to track overall business success. To find development prospects and in order to assist with revenue trends, financial planning, preferred payment methods, as well as product category contributions are examined.

BO2: Delivery & Satisfaction

The main goal of this analysis is to assess logistical performance along with customer satisfaction. Several metrics including, on-time delivery rates, average delivery days, delivery status distributions and late delivery percentages are used to evaluate the efficiency of deliveries. To find possible operational problems, consumer happiness is further assessed utilizing review scores along with delivery performance trends.

BO3: Seller Performance

The analysis aims to evaluate sellers efficacy and the market contribution. It will assist in enabling managers to identify the highest performing sellers, track the revenue concentration, and support seller growth goals by classifying sellers into performance categories based on revenue generation and order volume.

BO4: Marketing Funnel

The analysis mainly focuses on assessing the efficiency of lead generation and conversion across various channels of acquisition. The dashboard will help to determine the best channels for customer acquisition, and offers insights into marketing effectiveness by tracking conversion rates by origin and analyzing lead progression through the sales funnel.

BO5: Customer Segmentation

The analysis uses revenue forecasting, churn risk assessment, customer lifetime value analysis and customer segmentation to assist in customer retention along with long term revenue growth. While predictive analytics approaches are utilized to highlight customers at high risk of churn and estimate future business performance, behavioral segmentation of customers are utilized to support targeted marketing efforts.

Collectively, these five objectives provide a comprehensive assessment of business performance and support data-driven decision-making across operational and strategic areas of the organization. When these objectives are considered together, they provide a relatively comprehensive outlook of organizational performance and also enable data driven decision making in operations, finance, marketing, and customer-centric departments.

1.5 Dataset Explanation

Three datasets have been utilised in order to build the BI solution.

- **Dataset 1:** The primary dataset is from Olist's Brazilian E-Commerce Public Dataset on Kaggle, which is one of the most extensively researched e-commerce datasets in both commercial and academic settings. A total of 112,650 transaction records ranging between 2016 and 2018 are included within the dataset, which comprises variables like order_id, customer_id, order_status, delivery timestamps, purchase_timestamp, review_score, and payment_value. The main fact table for the model is in this dataset. The main reason for using this dataset was due to the fact that it is regarded as considerably reputed since it has been utilized extensively in academic research and data science contexts, as well as business analytics initiatives making it a well known benchmark for e-commerce analytics.
- **Dataset 2:** The second dataset is the Olist Marketing Funnel Dataset, which was also obtained from the same publisher on Kaggle. It contains information related to lead generation, sales conversion, and seller performance, supporting seller segmentation and

business performance analysis. This dataset was used as can be considered a quite trustworthy resource since it was published by the same Olist data source as the core transactional dataset, enabling consistency in the data structure, business context along with seller related IDs needed for integration.

- **Dataset 3:** The final dataset is an AI-generated Customer Behaviour Dataset developed using ChatGPT and linked to the primary dataset through customer_id as the shared unique identifier. Variables consisting of estimated_customer_lifetime_value, digital_engagement_score, discount_sensitivity_score, customer_satisfaction_index, return_probability, promotion_response_rate, preferred_product_category, and payment_preference are included within this dataset. The CLV segmentation along with the churn risk analysis capabilities of the model are improved by this dataset. Despite being AI generated, the dataset was created utilizing business rules as well as realistic customer behavior variables to supplement the transactional data. It has been designed to assist with the analytical experimentation and predictive dashboard development rather than representing actual customer records, and as such, its limitations are acknowledged within this study.

1.6 Purpose of BI solution

The main purpose of the BI solution is to turn Olist's raw data into combined, useful insights that helps with operational as well as strategic decision making. The BI solution further seeks to give the relevant management personnel a comprehensive perspective of logistical efficiency, consumer behavior trends, seller productivity and revenue performance via interactive Power BI dashboards, allowing Olist to detect gaps in performance, efficient allocation of resources, make strategic decisions based on actionable evidence in a relatively competitive e-commerce environment by utilizing DAX measurements, KPI visualization and predictive forecasting.

2. Literature Review & Research Discussion

2.1 Comparison

2.1.1 BI systems in decision-making

Business Intelligence (BI) systems play an important role in supporting organizational decision-making by transforming raw data into meaningful and actionable insights. According to Yerpude (2023), BI systems create knowledge from information extracted from data collected

from multiple sources. Through data processing and analysis, organizations can convert large volumes of raw data into useful knowledge that supports strategic decision-making. This ability can help companies to gain a deeper understanding of their operations and make more informed choices.

Likewise, studies conducted by Eyadat (2024) show that business intelligence tools in Jordanian firms create a more effective decision making. These systems collect and analyse data from a range of sources, making it easier to get relevant data. Clearer picture of the situation allows for leaders to act on a more complete portrait of reality. Consequently, decisions are more well-informed and in practice reliable. There is evidence that such benefits go beyond the individual decision and result in improvements in the performance of an organization over time.

Judijanto (2024) took a wider look at 748 research papers over a wider time frame from 2000 to 2024. The study demonstrates that BI research has evolved from more basic subjects like analytics and big data to application-oriented ones like sustainable practices and workflow performance. Judijanto also lists some common issues that may hinder the implementation of BI, including data quality issues, system integration issues, resistance to change, and ethical concerns.

2.1.2 Data modelling techniques

A key aspect of the database and data warehouse development is data modeling, which is responsible for mapping real world entities, attributes and relationships to an information system. Various modeling techniques have been developed to support different data management needs, spanning from relational databases to more up-to-date Big Data environments.

On the basis of research carried out between 2013 and 2020, Santos (2024) traces the evolution of data modeling in the context of a linked information system. The review emphasizes the rising significance of graph databases and NoSQL modeling to help integrate and support connected data applications involving Big Data. An effective data model must not only capture the entities and relationships accurately, but also allow efficient processing of more and more complex and interconnected data sets, Santos said. Another important research gap identified in the study is the lack of research on reverse engineering of connected data models, indicating that there is still a need for further research in data modeling techniques for current data environments.

Ballantyne et al. (2020) stick with the old good days, when Entity Relationship (ER) modeling was the paradigm of relational databases. The authors offer a systematic methodology with steps that will help the database designer to move from the conceptual design to the logical

design using ER diagrams. They focus on identifying entity classes, attributes and relationships as the key concept of good database design. Despite the advent of new tools, Ballantyne asserts that this traditional approach is still effective. The study concludes the need for ER modelling as it gives a picture of business requirements prior to system implementation.

When observing the data that passes through the warehouse, Garani et al. (2012) combine star and snowflake designs. The "starnest schema" is introduced by Garani, extending the traditional fact-dimension schemas to nested dimensional schemas, where the dimensions are stacked on each other. This can enhance the performance of queries by accessing fewer tuples for analysis at query time, and by performing more efficient aggregations. The study demonstrates how to improve dimensional modeling techniques to enhance analytical processing and decision making applications.

2.1.3 ETL vs ELT research

The emergence of cloud based BI platforms has led to a relatively significant increase in attention to the ongoing debate of Extract, Transform, and Load (ETL) versus Extract, Load, and Transform (ELT) approaches. Vassiliadis (2009), highlights the historical background of ETL approaches and suggests that conventional ETL mainly consists of converting data in a staging environment before it is loaded into the relevant data warehouse. On one hand, while it offers quality assurance and lineage tracing, it necessitates a substantial preparation framework. ELT on the other hand, uses the processing capability of the destination platform in order to add raw data straight into the relevant environment before applying transformations. According to Janssen et al. (2024), data lake and warehouse frameworks significantly support ELT by integrating transformation logic into the storage layer. This is supported by Rambabu et al. (2023), which provides an analysis of ETL and ELT in cloud data warehousing, emphasizing how ELT uses the processing capacity of cloud platforms in order to speed up transformations while ETL is still essential for corporate governance and compliance management. Plattner and Zeier (2012) further highlights how real time processing accelerated the transition to ELT. Collectively these sources attest that increase in the debate of ETL and ELT approaches are due to emergence of cloud based BI platforms.

Within its data processing architecture, Microsoft Power BI supports both ETL and ELT techniques. While DAX calculated columns and measures enable in model transformations after data has been loaded, essentially acting as an ELT layer, Power Query Editor serves as the ETL

transformation engine, allowing data cleansing, reshaping, and preprocessing before model loading (Nabil et al., 2023). Power BI is especially well suited for complex analytical scenarios where computational flexibility and data quality needs must be matched simultaneously because it can support both inside a single platform. In a comparative study of ETL and ELT approaches in retail analytics settings, Rambabu et al. (2023) found that neither strategy is always the best, and a better strategy would be hybrid implementations that combine in model computation for analytical flexibility with preload transformation for data quality assurance which consistently produce better performance results.

2.1.4 Dashboard analytics effectiveness

Business Intelligence dashboards have long been known in literature for its usefulness in assisting businesses in their decision making. After conducting a relatively thorough analysis on dashboard research, Sarikaya et al. (2018) came up with a framework wherein they categorized the dashboards based on how interactive it was, design objective as well as usage patterns. The results of this framework suggests that creating dashboards that offer proper information exploration along with a higher level of engagement are of vital importance. The authors also highlight that one of the most important components of a successful dashboard is its ability to enable its users to analyze data at different levels of detail by utilizing interactive elements.

As such, BI dashboards can prove to be one of the most important tools for decision making especially in e-commerce settings. Prakoso and Nuryuliani (2019), highlights that BI tools' ability to convert operational data into useful information for strategic as well as operational decision making is especially important when it comes to retail settings where there is a need to efficiently and effectively analyze a significant amount of transactional data. When it comes to ways in which BI systems influence logistics decision making, Grabińska and Ziora (2019) highlights that managerial dashboard design, supply chain optimization and business process enhancement are among the most notable. Hence, these findings offer compelling evidence for the four dashboards created in this project to include customer satisfaction, delivery performance, revenue monitoring, and operational KPI tracking as key factors.

2.2 Identify research gap

While existing literature and studies provide a decent theoretical foundation for BI systems, data modelling and ELT/ETL processes, there still remains gaps in the research where these

concepts are applied to specific environments and industries, in this case marketplace-based e-commerce. Firstly, existing research on BI is mostly generic, and not industry-specific. These studies focus on generic organizational settings and not platform-based business models, where data is fragmented and data ownership is shared among sellers, customers and logistics providers. This limits the practical applicability of these frameworks.

Secondly, ETL/ELT research has been focusing on enterprise scale cloud setups, which suggests a gap in research on small to mid scale environments that feature hybrid workflows combining tools such as Power BI and Python together. Next, existing data modelling research does not address situations where synthetic and AI-generated datasets are integrated together with real world data, an increasingly common practice in this day and age, leading to a growing challenge to solve. Apart from that, the concept of integrating predictive analytics with BI dashboards is largely unexplored. This is because existing studies focus on predictive analytics and BI dashboards as separate disciplines. Limited research has been conducted on how predictive analytics such as churn risk scoring and customer lifetime value estimation can be embedded directly into interactive BI dashboards for non-technical business users at the operational level.

2.3 Position study contribution

This study intends to address the identified gaps in research through the design and implementation of an integrated BI solution for the Olist e-commerce platform using Power BI. This will demonstrate the use of BI in a real marketplace context, showing how Power BI addresses fragmented, multi-domain data in a marketplace environment. Next, the implementation of a hybrid ETL /ELT workflow combining Python for preprocessing and Power BI for transformations, showcases a replicable methodology for small to mid-scale projects that do not fall into enterprise-scale frameworks. Apart from that, the integration of an AI-generated dataset alongside real world transactional data into a galaxy schema data model expands existing data modelling practices, demonstrating how synthetic data can supplement real world transactional datasets where behavioral data is not available. Furthermore, this study contributes to Power BI and predictive analytics research by combining predictive analytics with Power BI dashboards, embedding churn risk scoring, customer lifetime value segmentation and revenue forecasting into interactive, easy to understand and accessible Power BI dashboards for non-technical business users, narrowing the gap between predictive modelling and analytics and operational decision making.

3. Methodology

3.1 Power BI workflow

The project was designed to be a structured Business Intelligence (BI) process, leveraging Microsoft Power BI to convert raw data into valuable business insights. The workflow started by gathering data from three main sources: Olist Brazilian E-commerce dataset, Olist Marketing Funnel dataset and an AI generated dataset of customer behaviour. These data sets were chosen to enable analysis of various facets of sales, customer satisfaction, salesperson performance, marketing effectiveness and customer segmentation in a single BI application.

The first step was data preparation done in Python, where data cleaning and data preprocessing were done prior to loading the data into Power BI. Values missing from the data were treated according to the type of the variable. For numerical attributes, median imputation was used. Replacement of values like "Unknown" or "No Review" were used for categorical field. A data quality check, duplicate detection and outlier analysis were also performed to look for data quality problems that might impact the integrity of further analysis.

Once the pre-processing phase was finished, the processed data was exported as CSV files and loaded into Power BI. A number of transformations were carried out within Power Query, including validation, standardisation and removal of the final duplicate data if needed. Then the datasets were mapped to a common data model using galaxy schema (dimension modelling and relationship mapping).

After the modeling phase, DAX measures were created to provide key performance indicators (KPIs) and enable business analysis. These resulted in interactive dashboards and visualisations of sales trends, customer satisfaction, seller performance, marketing conversion and customer behaviour. The final dashboard offers users numerous opportunities to delve into business performance with various perspectives. It makes analytics easy for all to access and use, without requiring additional support, which enables data-driven decisions.

3.2 Data integration approach

The data was drawn from various business segments, so it was necessary to combine them into a single system for analysing and making decisions. This was a one-to-one link that enabled easy access to operations, marketing, and customers.

The Olist Brazilian E-commerce dataset was used as the main dataset for transactions, which included records of customers, orders, products, payments, reviews, sellers and geolocation. This

data was the starting point for analyzing the performance of sales, consumer satisfaction, delivery efficiency, and the performance of sellers. The Olist Marketing Funnel dataset combined with transactional data, allowed us to assess what marketing efforts and activities were helping the company acquire and convert customers. An AI-generated customer behaviour dataset was also added to deepen the analysis, which includes segments of customer behaviour, attributes of customer loyalty and churn risk, customer lifetime value, spending patterns and more.

During the integration of datasets, common identifiers were used to establish relationships between datasets. The `customer_id` field was used to make sure that customer-level behavioural data is analysed together with the customer's transactional behaviour, as the customer is linked in the Olist dataset with the AI generated customer behaviour dataset. The `seller_id` was then used to link the seller information and the marketing funnel information, which helped to analyze the performance of the sellers' acquisition and execution. The primary Foreign Key in the Olist dataset used for establishing relationships between order, payment, review, and order item data is `order_id` while `product_id` was used for establishing relationships between product information and transaction data.

The aim of the integration process was to create a single data model for cross-functional business analysis. Being able to combine data from the transactional side and marketing and customers' behaviour enables a much more detailed view of the business performance than any of these data sets would have provided individually. This enables customers to have a single Power BI environment to see sales trends, customer satisfaction, the performance of the sellers, the effectiveness of the marketing and customer segmentation – all of which helps to make more informed, consistent decisions across business functions.

3.3 ETL/ELT implementation

Data cleaning is considered a key step as it has a direct impact on the quality of the KPIs, as well as the overall analysis and the output of the dashboards. However, to achieve consistency of the three datasets, a hybrid ETL/ELT technique was introduced to standardise, validate and integrate the data before the analysis.

In the extraction step, data was taken from Olist E-commerce data, Olist Marketing Funnel data and an AI generated customer behaviour data. The missing value was addressed by performing the median imputation and the missing categorical value was replaced by "Unknown" or "No Review" in Python. Other data pre-processing steps involved fusion of product category

translations, standardisation of city names to minimise inconsistencies, and categorisation of customer reviews as Positive, Neutral, Negative, and No Review for analysing customer satisfaction.

Data validation included ensuring the integrity of key identifiers like `customer_id`, `seller_id`, `product_id`, `order_id`, and `mql_id` — these are necessary for modelling relationships and for data integration. Many outliers were observed but no one was eliminated since they were the business activity. One record from closed deal dataset has an extremely high declared revenue; however, the revenue could not be matched with the related dataset. So that one record was deleted to avoid avoid skewing on later analysis. After the cleaning was completed, the cleaned datasets were exported as CSV files and were loaded into Power BI where the final cleaning was validated and consistency checks were performed using Power Query.

Power BI created calculated columns, tables, and measures in an ELT process, based on the loaded data. This enabled the creation of business metrics like YTD Revenue, YoY Growth, Late Delivery %, Seller Ranking, and CLV segmentation without relying on the source data. The hybrid ETL/ELT workflow was chosen due to the benefits of both approaches. Data quality issues were resolved before analysis with ETL processes written in Python and Power Query, helping to minimize the potential for inaccurate reporting. Then came Power BI with the ELT logic and that's when the calculations got seamlessly adjusted with users' changes to their dashboards. A pure ETL approach would be very inflexible for reporting and a pure ELT approach would make it more likely that low quality data would be fed into the analytical model. The combination of both styles allowed for a space to keep datasets clean and ensure that they were not compromised by features of responsiveness required for exploration. This middle ground was close to the overall analytics tasks that were required.

3.4 Data modelling & analytics process

A data model was created using Power BI to enable analytical reporting and interactivity to visualise the data on a dashboard, after applying the ETL and ELT processes. The final schema implemented was a Galaxy Schema also known as Fact Constellation Schema which includes multiple fact tables linked by the sharing of dimension tables. This is chosen due to the fact that the project brings together numerous business processes such as ecommerce deals, order fulfillment tasks, customer actions analysis, and marketing funnel measurement. Each process has its own fact table and shares common dimensions like customer, seller, product, and date.

Different dimensional modeling methods were assessed in the design process. If the business process is the primary focus of the analysis and there was only one main fact table, it would have been appropriate to have used a Star Schema. A Snowflake Schema provides a higher level of normalization by breaking the dimensions into more dimension sub-tables that can help to eliminate redundant data, but can also make the model more complex and the number of joins that need to be used for analysis more complicated. Both options offered lack the flexibility that is required for integration of multiple business processes into a single analytical environment, as a Galaxy Schema would. The Galaxy is a schema that enables multiple fact tables to align on the same dimensions, while preserving the simplicity and scaling capabilities of the schema.

The model was expanded with several hierarchies to improve the ability to analyze the results. Drill down to Year to Quarter, Month and Day analysis with date hierarchies to view performance over various time periods. Geographical hierarchies were also added to allow for analysis by State, City and Customer location. These hierarchical relationships enable easy navigation in reports and enable them to switch from a summary to a detailed view of business performance.

The cross-filtering was primarily intended for one way filtering from dimension tables to fact tables. The design enables filtering of the relevant transactional records as soon as the categories, such as Date, Customer, Product, Seller and Geography, are selected, and filters the path. This is a best practice in Power BI dimensional modeling, enhances query performance and aids in getting consistent analytical results throughout the dashboard pages.

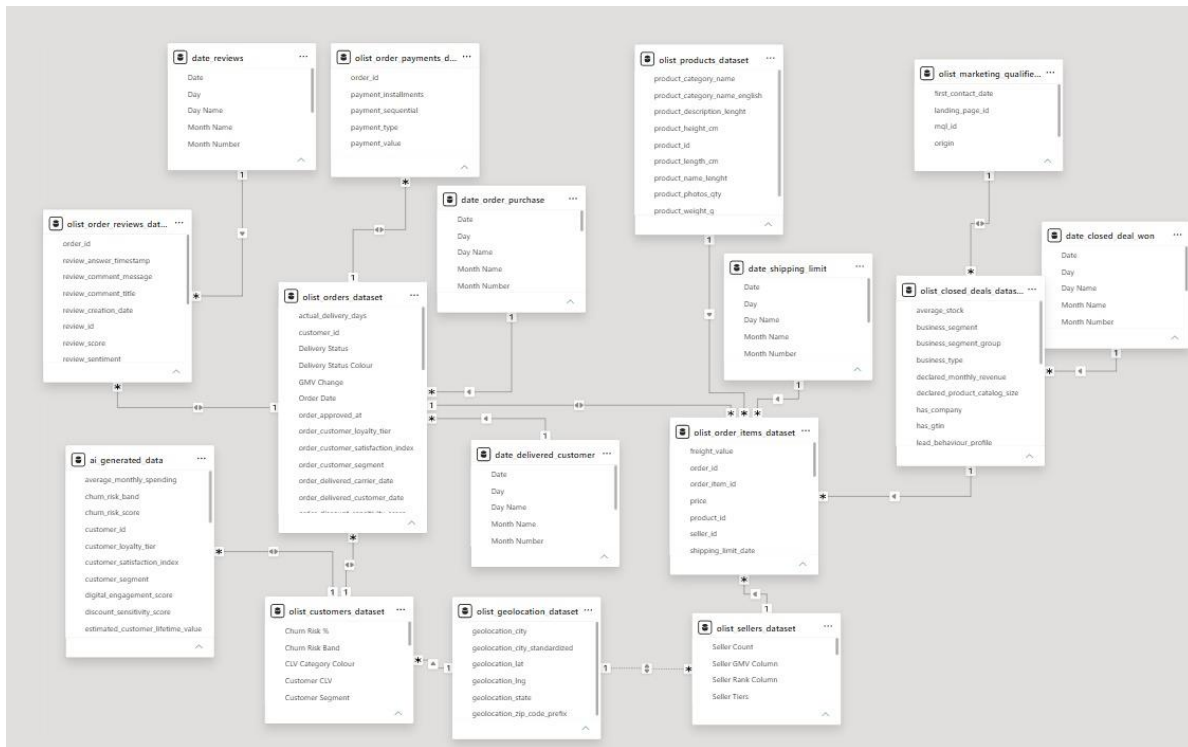


Figure 1 Galaxy Schema Data Model in Power BI

The final data model comprised sixteen tables, which were based on the Olist E-commerce dataset, the Olist Marketing Funnel dataset and the AI generated customer behaviour dataset. The model was quite basic and comprised three primary fact tables (Orders, Order Items and Closed Deals), and a set of conformed Dimension tables. This structure led to the establishment of a galaxy schema which enabled the integration of the analysis of sales performance, logistics operations, customer behaviour and marketing activities. The tables included were dimension tables like Customers, Products, Sellers, Reviews, Payments, Geolocation, Marketing Qualified Leads, and Customer Behavior. These dimensions in turn provided context to the filtering, segmentation, and in-depth analysis from several perspectives of business.

Five date dimension tables were added to the model to enable time based reporting. Every date dimension was associated with a particular line of business, such as order purchases, product deliveries, customer reviews, shipping activities and closed-deal transactions. Dedicated date dimensions enabled each process to be analysed on the date dimension that applied to it, without having to adhere to any common date dimension. This design also allowed for the examination of trend, seasonality and performance shifts throughout the various stages of the business process.

Shared identifiers such as `order_id`, `customer_id`, `seller_id`, `product_id` and `mql_id` were used to create relationships between the tables. These relationships enabled data from multiple sources to be connected and analysed within a single business intelligence environment. The galaxy schema made it possible to organize the data so that there was minimal data redundancy and yet sufficient flexibility for analysis. This also improved the performance of the dashboard since Power BI can process queries efficiently between different business processes.

The data model was mainly built to include one to many relationships, which are best for transactional analysis. For instance, a customer can place multiple orders, a seller can fulfill multiple order items, and a product may be part of many orders. Multiple order items and payment records are also tied to each order, which mirrors the real-world processes in the business. There were a small number of one-to-one relationships included where there was a corresponding relationship between the records, such as between each individual customer and their behavioural profile within the AI created dataset. Many-to-many relationships have been deliberately excluded as they can create confusing filter paths and impact model performance. Rather, other tables were created to connect Orders, Products and Sellers – these tables are called transactional tables. This method helped to keep the model structure simple, remain analytical and execute queries efficiently.

Lastly, calculated measures and KPIs were created according to the five business objectives: Sales performance, Customer satisfaction, Seller effectiveness, Marketing conversion and Customer behaviour (CLV, Churn risk, Loyalty metrics) (Section 4.2) using DAX. These measures provided the foundation for the visualizations on the dashboard, such as KPI cards, line and bar charts, maps, funnel charts, and customer segmentation. Interactive filtering and drill-downs were added to all dashboards, enabling users to navigate from summary to detail levels across time periods, product categories, sellers, geographic regions or customer segments. In this way, the dashboards could be used to monitor strategic goals as well as to make operational decisions in a single analytical environment.

3.5 Business Process Analysis

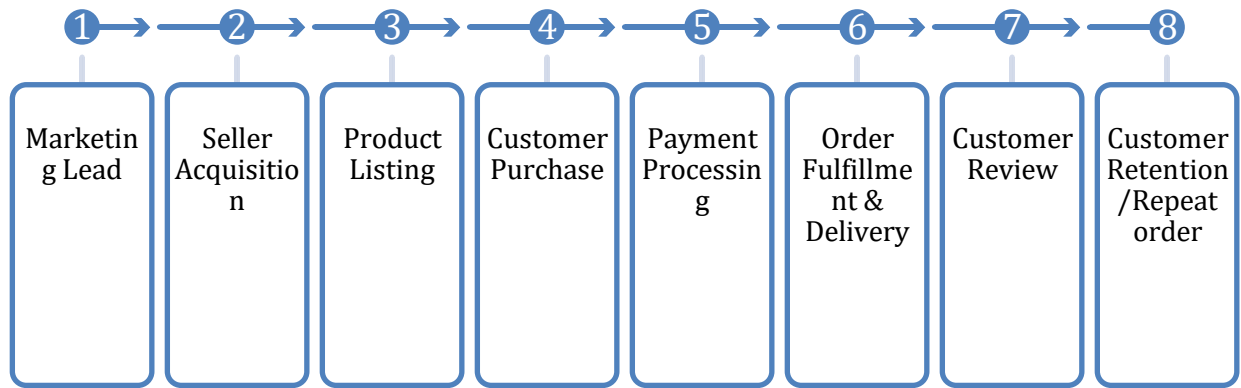


Figure 2 Olist Marketplace Business Process Workflow

Olist's main business process starts with acquiring sellers via the marketing funnel, where sellers become marketing qualified leads (MQLs) and eventually marketplace sellers. Onboarded sellers list products that are added to the marketplace for customers to purchase. The customers order, pay and receive products via third-party logistics providers. After the delivery, customers can leave reviews and ratings that affect the sellers' performance and customer satisfaction. Last but not least, customer behaviour and purchase history helps to retain and repeat purchases activities, closing the business cycle.

Before the BI solution was implemented, there were several disconnected sets of data that contained information on marketing activity, seller performance, customer transactions, logistical operations, and customer behaviour. This presented a number of operational challenges such as poor visibility of delivery performance, challenges in assessing seller contribution, customer data disassociation and lack of end-to-end business performance measurement. Consequently, management had to make operational and strategic decisions relying on different reports and manual analysis.

The Power BI solution helps overcome these challenges by connecting data from all the business processes to one analytical environment. Decision makers are able to evaluate the seller performance, monitor the logistics efficiency, analyse the marketing effectiveness, understand customer behaviour or find operational bottlenecks at various stages of the marketplace workflow through interactive dashboards, KPI monitoring and predictive analytics. This enhances the organizational visibility and enables the better utilization of data for making decisions in various business processes.

4. Results & Analysis

4.1 Dashboard interpretation

Variable	Mean	Median	Mode	SD	Variance	Skewness
Payment Value	154.10	100.00	50.00	217.49	47,303.67	9.25
Review Score	4.09	5.00	5.00	1.35	1.82	-1.36
Customer Lifetime Value (CLV)	8,139.38	1,803.14	50.00	17,941.53	321,898,586.40	4.19

Table 1 Descriptive Statistics of Key Business Variables

The above table shows a summary of the distribution of major business variables that were identified when analyzing descriptive statistics. Payment Value was significantly right skewed, in that the skewness value at 9.25, and a mean of 154.10 compared with a mode of 50.00. This indicates that most transactions are relatively small, but a number of high value purchases increase the average overall. Review Score, on the other hand, was left skewed with a skewness of -1.36, and an equal mode and mean of 5.00, suggesting that customer satisfaction is generally high. Lastly, Customer Lifetime Value (CLV) shows a substantial variation with a mean of 8,139.38 that is considerably higher than the median of 1,803.14, and the distribution is positively skewed at 4.19, indicating that a relatively small group of customers contributes to a disproportionate share of the total CLV. All in all, these values support the need for customer segmentation, seller performance analysis, and targeted retention strategies within BI Solution.

4.1.1 Executive Dashboard: Sales and Revenue Performance (BO1)

Olist connects independent sellers with consumers across Brazil's 27 states, supporting activities that range from seller acquisition and order fulfilment to transactions and customer feedback. Prior to this project, transactional, marketing-funnel, and customer behavioural data existed in separate datasets, limiting the ability to analyse performance across business functions. To bridge this, a set of 4 dashboards was created, which is based on a 16-table galaxy schema, that combines all three datasets in a single analytical environment. The cross-functional analysis is done based on this consolidated model and according to the project's five business objectives.



Figure 3 Executive Dashboard

The Executive Dashboard is created using seven KPI cards and eight visualizations in one screen, which shows Olist's commercial performance from Sept 2016 to Oct 2018. During this time, total GMV has amounted to R15.82 million while total orders have totalled 98,660 representing about 113,000 products. The Year onYear (Y0Y) Growth rate of 207.63% indicates that there was substantial business expansion overtime, and is further supported by The Year-over-Year (YoY) Growth rate of 207.63% indicates substantial business expansion over time. This growth is also reflected in the Revenue Growth chart, which shows that the rise in Total GMV is very similar to the rise in Total Orders. The revenue growth trend indicates that the platform experienced increased transaction volumes, rather than a rise in the average size of orders, thus suggesting that revenue growth was attributable to wider customer buying activity.

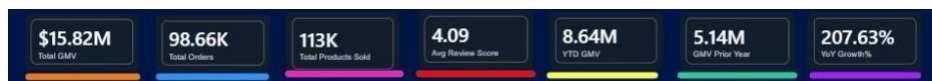


Figure 4 Executive Dashboard KPI cards



Figure 5 Revenue momentum combo

The MoM Growth and Total Orders chart shows a sustained increase in GMV from late 2016 to mid-2018, with the sharpest peak occurring in November 2017, which aligning with Black Friday, wherein customer purchase intent grows in general, thus, highlighting the strong influence of seasonal demand on Olist's revenue. This provides Olist's with insights for designing future marketing campaigns, inventory planning and logistics resource allocation for future seasonal peaks. Following this, growth rates became more stable during 2018, before turning negative by September 2018. However, because the dataset ends at October 2018, this decline may reflect the effect of a truncated observation period over, a genuine slow down in revenue. Additional data beyond 2018 is needed to conclude the reasoning behind the sharp decline.

The Revenue by Payment type chart signals to the management that there could is a risk of concentration within payment behaviour with credit card accounts dominating the donut chart at 78.09% of total payment value, in comparison to boleto at 17.51% and debit cards and vouchers together accounting for 4.58%. This dependence on a single payment channel is a cause of concern as any disruptions to credit card processing or regulations would disproportionately affect revenue. It is vital that Olist's product team look into incentivising the usage of boleto and debit card for payments to reduce the concentration risk.



Figure 6 Payment Type Donut

The Total GMV by Product Category treemap and Average CLV by Product Category chart provide complementary insights on product category performance. Purely based on GMV, Health and Beauty tops the charts at 1.44 million, followed by Watches and Gifts, and Bed, Bath and Table as the strongest performing product categories. Therefore, it is important to maintain seller acquisition and product availability for these categories to sustain the strong sales performance.

The CLV chart, on the other hand provides a varying perspective wherein other categories that do not contribute a huge percentage of GMV, such as the Small Appliances category, still record the highest average customer lifetime value. This indicated that customer purchasing from these categories exhibit stronger repeat purchase behaviour and generate greater long term value. This highlights the importance for Olist's team to not only evaluate product categories on current sales contribution, but to have a forward looking insight as well. Despite lower GMV, categories with higher CLV represent valuable opportunities for customer retention and long term, stable revenue growth. Overall, Olist's category investment decisions should balance both short-term revenue performance with customer lifetime value so as to ensure that both immediate and future business outcomes are considered.



Figure 7 Delivery status pie chart (left); Revenue vs Churn line (right)

The Order Count by Delivery Status pie chart shows that most orders were delivered on time at 89.15%, followed by 7.87% which was delivered late, and 2.98% which were not delivered. This provides important context for the overall picture wherein, as GMV is growing steadily throughout the period, churn risk is also growing. It can be understood that the platform is successfully generating more sales, but it is also facing growing challenges when it comes to customer retention. As a result, the positive year-on-year growth should not be viewed as the only positive indicator of business, because despite strong platform growth, rising churn risk also suggests that customer retention and operational performance is not improving at the same pace. This is a long term risk and could indicate an ongoing issue for Olist because without a steady stream of new customers, the company will also need to keep its current customers. *(This is the total on-time rate, which is calculated by taking all orders over time, and is different from the KPI that appears on the Logistics Dashboard, which includes only delivered orders (91.9%).)*

4.1.2 Operational Dashboard: Seller Performance and Marketing Funnel (BO2 and BO4)



Figure 8 Operational Dashboard

By consolidating seller performance and marketing effectiveness into one Operational Dashboard, BO2 and BO4 are now addressed. The overall business indicators are quite good, but

a closer look at the performance of each seller shows there are some structural imbalances not captured by these overall indicators.

Seller Tier	Count of Seller Count	Total Orders	Seller GMV	GMV % of Total
Platinum	60	28833	5,108,502.07	32.29%
Gold	150	23318	3,983,339.07	25.17%
Silver	300	20532	3,175,275.58	20.07%
Bronze	2585	28779	3,555,853.38	22.47%
Total	3095	83862	15,822,975.08	100.00%

Figure 9 Seller tier matrix

As shown in the above matrix, a small number of sellers that perform better earn most of the revenues, comprising 60 sellers out of 3,095 sellers, representing 32.29% of total GMV (R\$5.11 million). The GMV vs. Seller Rank by Tier scatter plot also reveals that the distribution of revenue is highly imbalanced with few users making up a significant portion of the total GMV of Olist's sales.

The problem with this concentration is that if some of the Platinum sellers slow down, it would have a disproportionate effect on overall activity performance for Olist, which could create a business risk. Thus, it is critical to create the seller success programmes to attract not just high-performing sellers but to expand lower-level seller with great increase capacity. Together, the Bronze and the Silver tier represent a total GMV of R\$3.56 million, indicating that there is still potential for this segment to be underdeveloped and can provide additional growth potential for Olist, beyond just increasing the number of sellers in the Platinum tier. Olist can also flag high growth Bronze and Silver sellers and offer focused assistance to boost long term marketplace performance more broadly, to mitigate concentration risk.



Figure 10 Lead progression funnel

Following the existing pattern found, the market analysis also reveals substantial variation across acquisition channels that is masked when looking at the overall strong performance. The Lead Progression through Sales Stages funnel indicates an effective lead conversion at the aggregate level, with an overall conversion rate of 10.5%.

However, further channel-level analysis reveals a more varied view of marketing performance. Organic search is the strongest acquisition channel, achieving a conversion rate of 3.4%, while also demonstrating shorter sales cycles and stronger downstream seller GMV. Social media, on the other hand, records a conversion rate of below 1%, while requiring the longest average time to close a deal. These differences are further shown in the Avg Days to Close and Seller GMV by Origin chart, which highlights the relationship between origin, conversion efficiency and seller value.

Overall, it can be concluded that there are inefficiencies in the underperforming channels. Based on this finding, Olist should shift marketing investments from the lower performing channels to channels with higher returns such as organic search, leading to an improvement in lead conversion efficiency and a higher quality of acquired sellers without having to further increase marketing expenditures.

It is important to note, that this dashboard also resolves a prior data management gap: the MQL and transactional datasets were previously disconnected, making channel-level acquisition cost uncomputable. The bridge join on `mql_id` makes it calculable for the first time, enabling true channel ROI comparison.

4.1.3 Logistics & Delivery Performance Dashboard: Delivery Efficiency and Customer Satisfaction (BO2)



Figure 11 Logistics Dashboard

In the context of logistics management, the operational insights to address BO2 through the Logistics and Delivery Performance Dashboard were obtained through converting delivery and ordering fulfillment data. At a glance, delivery performance seems strong but with a more in depth analysis certain variances are noted which are not visible through platform-wide metrics.

The On-Time Delivery gauge reports 91.9% and the Executive Dashboard reports 89.15%. This variance can be attributed to the Executive dashboard including undelivered orders which stand at 2.98% whilst the gauge excluded undelivered orders: $89.15\% \div (89.15\% + 7.87\%) = 91.9\%$. Both dashboards describe the same performance from different lenses, taking in account the different denominators with the gauge providing a delivered-orders-only view.



Figure 12 Actual vs estimated delivery time

One of the higher priority findings would be the variance between customer expectations and delivery performance, Figure 12 shows this where the estimated delivery window was 24.40 days which were delivered within an average of 12.50 days. During this study period, Olist has consistently exceeded its own estimated deliveries which are shown as delivery times declined steadily whilst delivery windows narrowed at a slower pace. The Delivery Performance Over Time chart supports this, showing a gradual increase in the on-time deliveries across a large volume of the dataset. Olist has an opportunity to review its delivery promises due to their actual delivery performance being substantially better than their estimate. Improving delivery estimates could help improve customer satisfaction and confidence without the need for additional investment.

The Top 10 States by Delivery Time and Late Delivery % by Customer State show that performance is not evenly distributed across regions, with Maranhão, Piauí and Alagoas being identified as the weakest performing states with a late-delivery rates reaching 20% compared to the average of 8%. States such as Roraima (RR) and Amapá (AP) fall within the high cost, slow-delivery quadrant citing carrier performance and logistical issues as compared to the southern counterparts that fall within the high-efficiency quadrant which demonstrates stronger delivery performance providing a practical benchmark for improvement in underperforming regions as explained by The State Logistics Efficiency Matrix.



Figure 13 State logistic efficiency matrix

Findings describe how delivery challenges are displayed in rather isolated regions instead of the entire platform, as a result of this network-wide investments may yield less benefits compared to targeted actions such as contract renegotiation, carrier performance reviews or alternative logistic arrangements which is supported by The Total Orders by Geolocation City and Delivery Status map. When comparing order volume, order majority is concentrated in the south-eastern regions like São Paulo which record strong on-time delivery as opposed to the northern and north-eastern regions which proportionally show a larger portion of late and undelivered orders. Considering what the visualizations describe, logistic issues appear to be primarily regional and should be addressed through targeted improvement initiatives.

The Average Review Score by Delivery Status chart shows the strong relationship between customer satisfaction and delivery performance with on-time deliveries receiving an average score of 4.3 compared to the 2.6 for late deliveries and a 1.8 for undelivered orders. This shows the polarizing difference between customer responses with 57.8% of reviews being 5 stars and 11.5% rated one star, The Review Score Distribution chart supports this which suggests that delivery outcomes tend to skew customers reports of a positive or negative experience.

The importance of delivery performance cannot be understated where the likelihood of receiving a one-star review is increased by 6.60 times for Not Delivered, with Late Delivery increasing by 5.98 times. Compared to the other factors explored in the analysis like freight value, payment method and customer segment, this seems to be the stronger factor as visualized by the Operational dashboard. This also makes it heavily relevant to the logistics analysis as it quantifies the direct impact between customer experience and delivery performance, with findings showing

that improvement of logistics reliability is more likely to have a positive effect on customer satisfaction as opposed to changes in product assortment, category mix or merchandising strategy.



Figure 14 Repeat purchase scatter

Expanding on customer satisfaction, customer retention is an important aspect to consider which is described in The Repeat Purchase Probability vs Average Delivery Days by Customer Segment scatter plot. This shows how High-Value customers and At-Risk customers are concentrated into segments with shorter delivery times and longer delivery times respectively. While this doesn't establish causality, it provides sufficient evidence on how customer retention outcomes are decided by delivery performance. Factors such as delivery speed and reliability also come into play when considering customer retention, likelihood of repeat purchases and long-term value. This results in delivery consistency being one of the key factors that have the potential to generate benefits across business objectives and drive revenue growth.

4.1.4 Predictive Dashboard: Customer Lifetime Value and Churn Risk (BO5)



Figure 15 Predictive dashboard

The Customer Intelligence and Predictive Analytics Dashboard addresses BO5 by integrating the AI generated customer behaviour dataset with the transactional data from the first dataset. This dashboard supports a more forward looking analysis by producing predictive insights when it comes to customer value, engagement patterns and churn risk. Before moving forward, it is important to consider the data quality during interpretation as dataset 3 is synthetically generated. Churn risk should thus be considered as a segmentation indicator rather than a validated prediction of Olist's customer base. Rather, the following insights aim to provide a general guidance over real customer behavior data before being used for operational retention decisions.

The Customer Churn risk gauge reports an average churn risk score of 89.16%, which generally looks like a concerning amount. However, this KPI needs to be examined alongside the Customer Segment distribution for a more well rounded business insight. The customer base is dominated by Regular and Occasional Customers accounting for almost 80% of the entire base. In contrast, High Value customers make up only 0.1% and New Customers make up 18.72%.

Given this distribution, the high aggregate churn risk score is likely due to the nature of online marketplace commerce wherein a large proportion of customers only make one or a small number of purchases before going back to being inactive.

This analysis aids Olist's management team in proving that the overall churn risk should be interpreted alongside customer segmentation for interpretation for business decisions. The important insight here lies in identifying and understanding which customers exhibit an elevated churn risk despite meaningful engagement and/or value. Based on the above, the At-Risk segment needs to be looked into as an opportunity where retention initiatives are likely to generate measurable business impact for Olist.



Figure 16 Churn gauge (left); Customer segment donut (right)



Figure 17 Revenue Forecast



Figure 18 Revenue Growth Drivers

The Revenue Forecast for the next 6 months chart utilises Power BI's smoothing model to project GMV for the next 6 months following October 2018, with a 95% confidence interval. While the forecast indicates continued revenue growth, the confidence interval widens over time, suggesting that future revenue estimates become less certain as the forecast extends and should therefore be interpreted as a range of possible outcomes rather than a single expected value.

The Revenue Growth Drivers waterfall Chart aids in breaking down the contribution to revenue in monthly periods. Aforementioned, November 2017 contains the sharpest peak due to Black Friday period, and there is a stable level of positive growth through January 2017 to 2018 before becoming more volatile.

To synthesize this, it can be confirmed that Olist's revenue growth is influenced by both long term expansion and peak seasonal events as well. Future financial planning should account for these seasonal fluctuations by looking at just the overall linear growth. The seasonal peak of Black Friday can be incorporated into forecasting and budgeting for this period, while lower bounds in the forecast confidence can act as a conservative reference point when planning efficiency in operational resources and financial commitments.

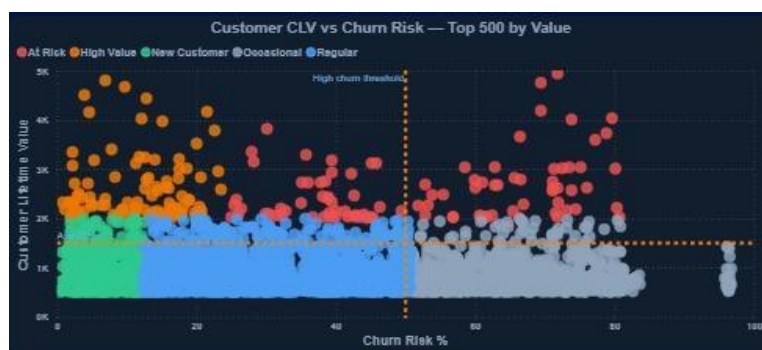


Figure 19 CLV vs Churn scatter

One of the most actionable views in the predictive dashboard is the scatter plot of Customer CLV vs Churn Risk: Top 500 by Value, as this analysis is a combination of the customer value and the risk of churn. High value customers tend to be grouped together in the high CLV, low churn risk quadrant of the chart and are seen to be both valuable and relatively stable. The main goal for this group is to help retain customers and keep them engaged in the program, rather than implementing intensive recovery efforts for all groups.

At-Risk customers, on the other hand, are mostly concentrated between BRL 2,000 and 5,000 CLV, with churn-risk scores exceeding 60%. These customers are the greatest potential for retention as they are still valued customers but are at a higher risk of being lost. With a conservative

assumption of 20% of these achieving retention through targeted retention efforts, the possible At-Risk average CLV of BRL 3,735 gives the value of the top 500 customers alone of BRL 64,900. This is a theoretical costing that was extrapolated from behavioural indicators in synthetic data, but it shows the potential cost of recognizing and attracting at-risk customers before they churn.

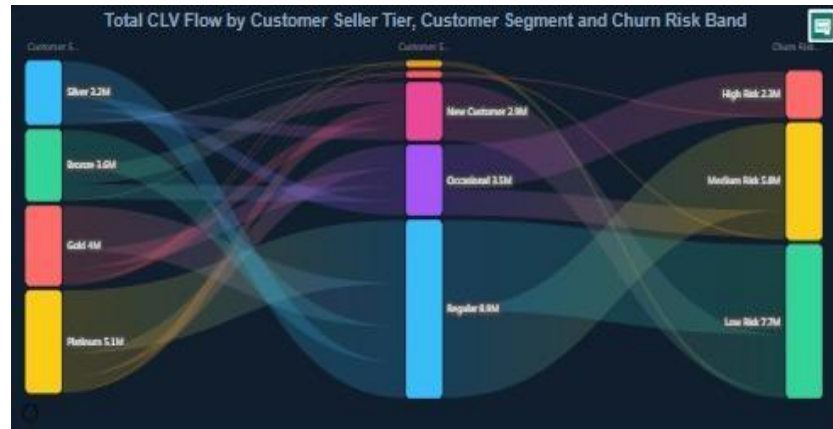


Figure 20 CLV Flow

The above Total CLV Flow Sankey diagram shows a cross-functional perspective of Customer Value as it moves from the Seller Tier to the Customer Segment and then to the Churn-risk Category. This visualization reveals a pattern not visible by the GMV analysis wherein, customer lifetime value for Platinum sellers primarily flows to the Low, and Medium-Risk churn categories, while the CLV for Bronze sellers is more spread out across all risk levels, including the High-Risk category. This shows that Platinum sellers are not only making more money, but also customers that have better loyalty traits. Therefore, the business value of having the top-performing Platinum sellers goes far beyond their impact on total GMV (32.29%) and also their contribution in creating more sustainable long-term customer value.



Figure 21 CLV Decomposition Tree

The Customer Lifetime Value Breakdown Decomposition Tree is a more detailed breakdown of the factors of CLV. By drilling down on loyalty tiers and customer segments, the visualization shows an increase in customer lifetime value from one tier to the next. This pattern indicates significant differences in value among customers within each loyalty tier, and that there is a huge jump when customers move from one tier to the other. These findings can help inform the design of retention and loyalty initiatives, which will help identify customers that are moving toward a new value tier. In addition to applying the same incentives to all customers, retention programs can be specifically aimed at customers who are most likely to hit important CLV milestones, which may enhance the efficiency and effectiveness of customer development programs.

4.2 KPI Framework and DAX Implementation

The BI Solution's analytical layer is a combination of three different metrics, each of which serves a separate role in the reporting environment. First and foremost, several descriptive fields were generated to be used in the Python based pre processing stage and then included in Power BI as prepared columns, including factors such as standardized city names, delivery delay indicators, along with rule-based emotion categories for reviews. Secondly, customer behavioural attributes fields from the synthetic DS3 dataset such as customer lifetime value (CLV), churn risk score, and customer segment were directly imported as source columns instead of being calculated within Power BI. Moreover, a combination of DAX measurements and computed columns were created in Power BI in order to aggregate, rank, as well as analyze the combined data. These measures altogether form the in model analytical layer that powers the dashboards. Figure 22 summarizes the core measures.

Each measure that was created was developed in order to address a relevant business question that could otherwise not be answered with the originally disconnected datasets. Total GMV, YTD GMV, as well as YoY Growth % are among the main revenue measures that assist in the analyzing the sales performance and business growth, while the Late Delivery % can be utilized to quantify the SLA breaches among the delivered orders, enabling logistics performance to be measured relatively consistently. Furthermore, in order to group sellers into Platinum, Gold, Silver and Bronze categories for the performance analysis, Seller GMV was utilized as a basis for Seller Rank and Seller Tier classifications. Additionally, a comparatively dense ranking was applied to the Seller Rank in order to make sure that the seller tier assignments were consistent no matter

what filters were applied within the dashboards. For the customer segmentation and retention analysis, Avg CLV and Churn Risk % has been used to surface the behavioural metrics provided by the DS3 dataset, while Lead Conversion Rate has been used to make comparisons across acquisition channels.

Furthermore, by utilizing the Power BI's Analytics Pane forecasting function, revenue forecasting was implemented, separate from the DAX layer, which enables a smoothing model to be applied to historical GMV data. This method is much better suited for time series data exhibiting trends and seasonal fluctuations as it places a significantly greater weight on recent observations. Since the same sixteen-table galaxy schema is utilized across all calculations, any filter that is applied within the dashboard will automatically propagate across all the relevant visualizations and measures. For example, when a Seller Tier is filtered, conversion, revenue and customer satisfaction are updated accordingly. This aids users in analysing business performance for decision making from different perspectives within a single environment.

Measure Name	DAX Formula (Full)	Business Purpose
Total GMV	Total GMV = SUM(olist_order_items_dataset[price]) + SUM(olist_order_items_dataset[freight_value])	Computes full transacted value including freight for BO1 revenue monitoring
Prior Year GMV	GMV_Prior_Year = CALCULATE([Total GMV], DATEADD(date_order_purchase[Date], -1, YEAR))	Computes GMV offset 12 months for YoY comparison
YTD GMV	YTD_GMV = TOTALYTD([Total GMV], date_order_purchase[Date])	Accumulates GMV from year start; resets each January for BO1 YTD tracking
YoY Growth %	YoY_Growth% = DIVIDE([Total GMV] - [GMV_Prior_Year], [GMV_Prior_Year])	Year-on-year revenue growth rate expressed as a percentage for BO1
Late Delivery %	Late Deliveries = CALCULATE(COUNTROWS(olist_orders_dataset, olist_orders_dataset[order_delivered_customer_date] > olist_orders_dataset[order_estimated_delivery_date], NOT ISBLANK(olist_orders_dataset[order_delivered_customer_date])) Late Delivery % = DIVIDE([Late Deliveries], [Delivered Orders])	Proportion of delivered orders arriving after estimated date for BO2 SLA monitoring
Seller GMV	Seller GMV = SUM(olist_order_items_dataset[price]) + SUM(olist_order_items_dataset[freight_value])	Aggregates revenue at seller level for BO3 tier ranking
Seller Tier	Seller Rank = RANKX(ALL(olist_sellers_dataset[seller_id]), [Seller GMV], DESC, DENSE) Seller Tier = SWITCH(TRUE(), [Seller Rank] <= 50, "Platinum", [Seller Rank] <= 200, "Gold", [Seller Rank] <= 500, "Silver", "Bronze")	Assigns dense revenue rank to each seller for BO3 tier classification Calculated column classifying sellers into four tiers by rank percentile for BO3
Lead Conversion Rate	Lead Conversion Rate = DIVIDE(COUNTROWS(olist_closed_deals_dataset), CALCULATE(COUNTROWS(olist_marketing_qualified_leads_dataset), ALL(olist_closed_deals_dataset)))	MQL-to-close conversion percentage for BO4 channel evaluation
Avg CLV	AVERAGE(ai_generated_data[estimated_customer_lifetime_value])	Mean projected customer lifetime value from DS3 for BO5 segment analysis
Churn Risk %	AVERAGE(ai_generated_data[churn_risk_score]) * 100	Platform-level churn risk score from DS3; sensitive to slicer context for BO5

Table 2 PowerBI DAX Measures and Business Purpose Mapping

4.3 Data-Driven Decision-Making Insights

The value of the BI platform is that it can relate metrics to specific business decisions, not the volume of information it shows. All dashboards share the same 16-table galaxy schema, which

also results in the same filters and slicers being applied to all related measure and visualizations creating static reports that become dynamic and interactive for use in decision making. This enables multi-disciplinary, and well informed decision making, for instance, the finance team can compare revenues for the current year with those years of the past, the logistics team can break down the individual states and review delivery time, freight charges, reliability of service and more; and the retention team can consider what groups of customers they have that vary in churn likelihood and in customer lifetime value (CLTV). All high level performance data is available to customers and then they can drill down into any of the topics to get more detail, without losing business context. As a result, decisions may be made based on summary metrics as well as based on operational and commercial factors which impact said summary metrics.

4.4 Implementation Challenges

There were a number of challenges encountered during development. Most notable was the integration of three datasets that have different structures, granularity, and business objectives. Creating reliable relationships based on `customer_id`, `seller_id`, `order_id`, `product_id`, and `mql_id` took a long time to assure that all records were correctly connected, and that cross-functional analysis was accurate. Data quality problems such as missing data, inconsistent categories, missing product categories and incomplete data also needed a significant amount of preprocessing before they could be modelled, thus the importance of data preparation for proper calculation of KPIs and reporting.

In dashboard design, a trade-off between analytical depth and usability emerged across four dashboards that focused on different business objectives. This was achieved by focusing on KPI cards, interactive filters, and drill-down instead of overcrowding with visuals. The other challenge was to embed predictive analytics in a traditional BI environment. There was a need to integrate transactional data, behavioural attributes and Power BI analytics into a single reporting framework to support revenue forecasting, CLV segmentation and churn-risk analysis. Solving these challenges led to a single analytical environment that could help with descriptive and predictive business analysis.

5. Conclusion & Recommendations

5.1 Summary of findings

The analysis showed that there was a considerably consistent picture across all the five business objectives in terms of the overall performance, however, significant risks and opportunities emerged when the data was explored in more depth at the platform level.

Olist's Sales and Revenue (BO1) showed a GMV of \$15.82 million with YoY growth at 207.63%, which indicates that there has been a significant business growth over the study period, even though about 78% of the payment value has been achieved by credit cards, meaning there is a relatively high dependency on a single payment channel. For Delivery and Satisfaction (BO2), the average delivery time was 12.5 days while the communicated estimate was 24.4 days meaning that the orders were being delivered on average before the communicated time. While this overall score is quite good, the percentages of late deliveries seen in multiple northern and north eastern states were higher than 20% and the delivery status was one of the main factors affecting customer satisfaction.

When it comes to Seller Performance (BO3), the revenue generation was limited to a few top sellers. It is quite remarkable that the 50 Platinum sellers contributed 32.29% of GMV while being only 1.62% of the total seller base, meaning that the platform has a very limited number of sellers. Marketing Performance (BO4) has a conversion rate of 10.5%, however, there was a significant difference between the conversion rates by acquisition channel. Organic search data clearly demonstrated that it performed better than social media in all metrics, including conversion efficiency, length of sales cycle and the value of the downstream seller, indicating there may be greater opportunities for budget allocation.

Lastly, when it comes to Customer Retention (BO5), marketplace platforms consistently had high numbers of one time and low frequency customers, which skewed churn metrics. The relationship between customer lifetime value and churn risk analysis proved more interesting. The highest CLV and high churn risk scores were those of customers between BRL 2,000 and BRL 5,000, and the top 500 customers had an estimated LTV of BRL 64,900.

5.2 Business implications

There are a number of implications for Olist and other marketplace platforms from these findings. To start with, data integration is a strategic capability in itself. This project brought

together datasets on transactional, marketing and customer behavior together for the first time, although they already existed before this, they were operating in isolation. Therefore, combining them into a single analytical model enabled cross functional analysis which then connected to marketing activity, customer behavior, seller performance and operational outcomes.

Furthermore, it is possible for an aggregate measure to mask some of the operational and strategic problems. The indicators for the platform include an on time delivery rate of 91.9%, an average of 4.09 on reviews and a lead conversion rate of 10.5% which are all comparatively good. However, further investigation has revealed that there are certain delivery deficiencies by region, that the income was focused in the hands of only a few sellers, and there was significant variation in the effectiveness of the marketing channels. The results highlight the need for metrics other than the summary to be analyzed for performance.

The analysis also revealed a high correlation between logistics performance and customer satisfaction. The average review ratings drop from 4.3 for on-time deliveries to 2.6 for late deliveries, suggesting that enhancing delivery reliability is likely to result in more benefits to customer satisfaction than many other operational initiatives.

Lastly, measures of marketing performance must be considered in relation to metrics other than conversion rates. Acquisition channel analysis revealed not only varied conversion rates, but also highlighted the different customer and seller value and retention rates for each channel. This indicates that customer lifetime value and retention results should be considered alongside the traditional conversion metrics for measuring marketing effectiveness.

5.3 Strategic recommendations

Based on our findings, we propose several strategic recommendations that can be made for Olist. Firstly, the estimated delivery times communicated at checkout should be changed from the current 24 day average to approximately 15 days. This would require no investment in the logistics department but would improve the experience for the customer right away. Next, we believe the logistics department should renegotiate the delivery terms with the carriers for the northern and northeastern states. These states fall within the quadrant of the State Logistics Efficiency Matrix that features the highest delivery times and costs. Changes to the carriers for these states would require a structural change to the current delivery terms for the company. Following that, the product team should create promotions and incentives that encourage customers who already use

credit cards as a payment method to opt for boleto or debit cards. This will hopefully help to mitigate the impact of credit card disruptions in the future.

Apart from that, the seller success team should implement a system that monitors the account health of the 50 Platinum sellers. The seller success team should also create a program that allows Bronze and Silver sellers who exhibit strong revenue growth to advance to higher selling tiers, thus reducing their likelihood of contributing to a concentration of sellers within high-risk categories in the future. Lastly, the marketing team should shift a part of the social media ad spend on organic search efforts. The Lead Conversion Rate by Origin analysis shows that organic search has an overall higher conversion rate (~3.4%) than social media channels, which are below 1%, while it also results in quicker deal closures and better seller performance. Overall, a boost in investment in the area of organic search would have a significant impact on the overall marketing efficiency and seller acquisition results.

5.4 Limitations and future improvements

There are a number of caveats in interpreting the results. First, the data set of customer behaviour (DS3) was created synthetically and therefore the results of CLV, churn-risk, and segmentation are directional and not an actual prediction of behaviour. These results need to be confirmed using actual customer behavioural and transactional data in future deployments. Secondly, the transactional dataset only covers ranging from September 2016 to October 2018. Growth towards the end of the analysis period could be an artefact of the data set boundaries and more recent data would support more valid trend analysis and forecasting.

Thirdly, the revenue forecast was done by an exponential smoothing model built in Power BI. Although the feature can detect historical trends and seasonality, it is not able to identify external factors, like promotion, macroeconomic factors, or competitions. These variables may help make better forecasts. Another restriction is the hybrid Python–Power BI workflow. It was very flexible in the early stages of development, but transformations were spread out in several tools. The greater centralization within Power Query, Dataflows or an enterprise ETL pipeline will provide greater maintainability and reproducibility. Further improvements may involve using existing real behavioural data to create a validated churn prediction model, clustering based on customer behaviour for segmentation, and facilitating automated data refresh capabilities for near real-time monitoring and decision making.

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